ASSESSMENT TOOL – 2

Course Code	CSA0702	**Course Title**	Computer Networks
CO Assessed	CO3	**Assessment**	Case Study
Weightage		**Total Marks**	

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Section / Batch	ECE-2024	**Date of Submission**	
Faculty Incharge	Dr R S Selvan	**Assessment**	Individual Submission

Total Marks	20	**Obtained Marks**	20

Code of Conduct:

10/8/26

I Cindy R / 192412571 *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Question - 1

Case: A cloud based video conferencing platform experiences 8% packet loss, 180 ms jitter, voice distortion, frozen videos frames, and delayed communication during peak hours.

Analyse whether the Problem is primarily caused by the transport protocol (TCP/UDP) or the application layer.

The issue is caused by both the transport protocol and the application layer, but it is primarily an application layer issue.

* Video conferencing applications generally use UDP because it supports real-time communication with low delay

* The reported 8% packet loss and 180ms jitter indicate unstable network conditions.

* Frozen video and delayed audio occur because the application's jitter buffer cannot compensate for excessive delay variations.

Therefore, the main problem is poor application-level buffering under adverse network conditions, while the underlying network conditions contribute to the issue.

Justify your answer based on the communication requirements of real-time multimedia applications.

Real time Multimedia applications require:

* Low latency (< 150 ms)

Low jitter (< 30 ms)

Minimal packet loss

Continous playback

UDP is preferred because

No connection setup delay

No retransmission

No acknowledgments

Lower protocols overhead

TCP is unsuitable because:

Lost packets are retransmitted

* Retransmissions introduce additional delay
* Congestion control reduces transmission speed.

Hence, UDP combined with adaptive application buffer provides the best performance.

Recommend suitable protocol-level or application-level Improve

Protocol-level Improvents

* Use UDP with RTP
* Implement RTCP
* Enable forward Error correction (FEC)
* Apply Quality of service (QoS) prioritization.

Expected Outcome

* Reduced delay
* Lower jitter
* Improved voice clarity, smooth video playback.

chniques Used

AnyCast DNS

* Multiple DNS server share one IP address
* Requests are routed to the nearest server
* Reduces latency significantly

DNS Pre-fetching

* Browser resolves commonly accessed domains before
* users click links.
* Eliminates lookup delay

TTL optimization

* Frequently accessed records : TTL = 300-600 seconds
* static records : TTL = 3600 seconds

Expected lookup Time :

* 800ms → Below 50ms

Evaluate ~~advantages~~ and limitations

Advantages

* Faster DNS resolution
* Better scalability
* High availability
* Automatic load balancing
* Reduced server load

Limitations :

* Higher deployment cost

Question 2

DNS lookup takes approximately 800ms, delaying webpa
loading - Case.

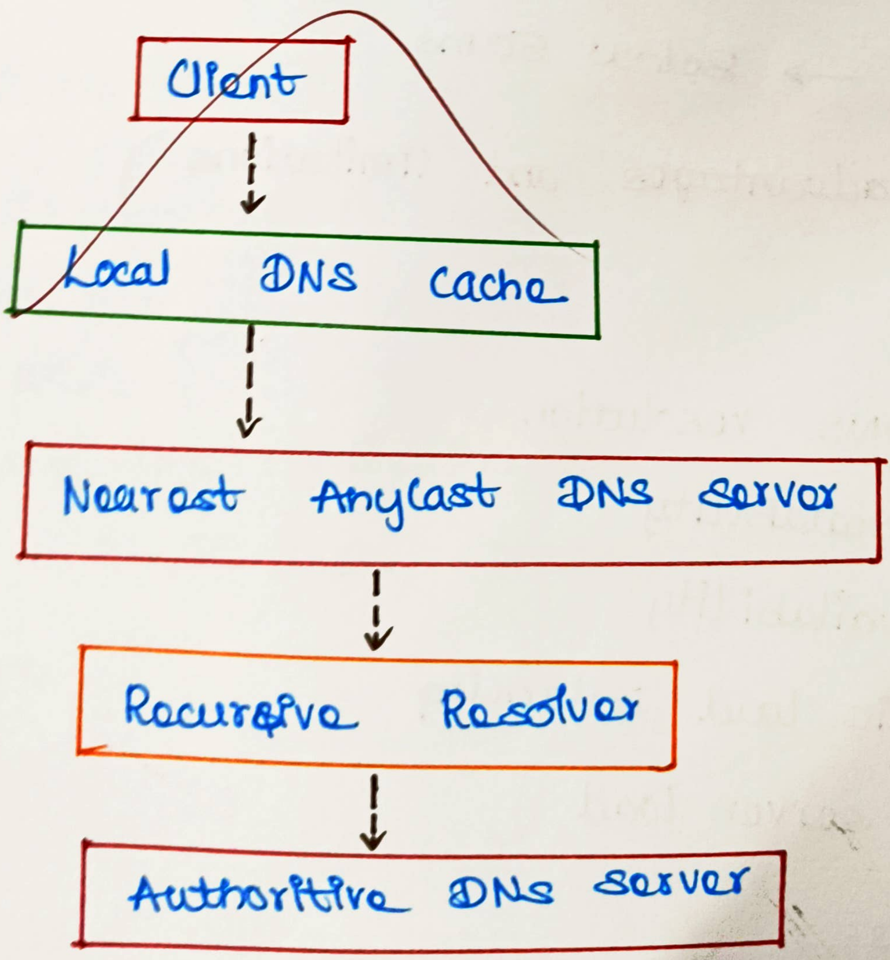
Analyze the reasons behind the high DNS resolution time

Possible cause include:

- single centralized DNS server
- Long physical distance between clients and DNS server
- DNS cache misses
* High DNS server workload
* Inefficient TTL configuration
* Network congestion.

These factors increase DNS query latency.

Propose a suitable DNS architecture



Features:

* Global Anycast deployment
* Geo - distributed DNS servers
* DNS load balancing
* Redundant authoritative servers
* Local caching

Benefits:

* Faster DNS lookup
* High availability
* Fault tolerance
* Automatic failover

Analyze TTL, Anycast routing, and DNS Caching

TTL (Time to Live)

Low TTL - Quick updates, Faster failover , More DNS traffic

High TTL - Better coaching, Lower DNS load, Slower updates

Anycast Routing:

Advantages:

* Routes to nearest DNS server
* Low Latency
* Automatic traffic distribution
* High availability.

* Complex Anycast routing
* Cache consisting challenges
* TTL tuning requires careful management

Question 8

Case : A cloud provider directs all DNS requests to a single primary server, resulting in 800ms DNS lookuptime

Examine the impact of centralized DNS architecture

Centralized DNS causes:

* High lookup latency
* Single point of failure
* Server Overload
* Poor scalability
* Increased network congestion
* slow application response

Result:

Users experience delayed webpage loading and poor service availability.

Design a distributed DNS solution

$\boxed{\text{Clients}}$ → $\boxed{\text{Nearest Anycast DNS}}$ → $\boxed{\text{Regional Recursive Resolver}}$ → $\boxed{\text{Multiple Authoritative DNS servers}}$

NS Caching

Benefits

* Eliminates repeated DNS queries
* Faster webpages loading
* Lower server workload

These techniques together improve performance, scalability, and fault tolerance.

Question 4

Case : A stock trading platform experiences acknowledgment latency increasing from 1 ms to 40 ms. TCP retransmissions, connection delays and buffering are observed.

Analyze three TCP-related factors responsible

a) Flow Control

TCP uses the receiver window to control transmission speed.

Problem:

If the receiver processed data slowly, the advertised window shrinks.

Effect:

* Sender pauses frequently
* Increased latency
* Reduced throughput

b) Nagle's Algorithm

Nagle combines small packets before transmission

Problem

stock trading requires immediate transmission of every order.

Effect:

- Artificial delay
- Increased acknowlegments time
- Higher transaction latency

TCP Factor	Effect on Trading Platform
Flow control	Delays order transmission due to reduced receive window
Nagle's algorithm	Delays and small buy/sell order packets, increasing acknowledgment latency
SYN Queue Overflow	Slows or drops new client connections during peak market activity

Overall Impact:

- Delayed trade confirmations
- Reduced transaction throughput
- Customer dissatisfication.

potential financial losses.

Outcome:

These TCP optimizations reduce acknowledgment latency 40ms toward 1ms, improve transaction reliability, minimize retransmissions and ensure efficient processing during peak trading periods.